Linear Algebra and Its Applications

Section 5.3: Diagonalization

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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5.3 Diagonalization

In this section diagonalization is over $\mathbb{R}$: only real eigenvectors can form the required basis of $\mathbb{R}^n$. Diagonalization is a change to an eigenvector coordinate system. If the columns of an invertible matrix P are eigenvectors and the corresponding eigenvalues are placed in the same order on the diagonal of D, then $AP = PD$ and $A = PDP^{-1}$. The requirement is *enough independent eigenvectors*, not merely enough vectors or roots of the characteristic polynomial.

The algebraic multiplicity of an eigenvalue is its number of appearances as a characteristic-polynomial root. Its geometric multiplicity is the dimension of its eigenspace. These are different notions. A repeated root must be checked by solving a homogeneous system; it neither guarantees nor forbids diagonalization.

Practice problem 5.3.P1: A high power from two eigenvectors

The characteristic polynomial is $(t-2)(t-1)$. Solving the two homogeneous systems gives eigenvectors $(3,2)$ for 2 and $(1,1)$ for 1. Thus

$$P = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}.$$

In $(PDP^{-1})^8$, every adjacent $P^{-1}P$ cancels to I, giving $A^8 = PD^8P^{-1}$. Since $D^8 = \operatorname{diag}(256, 1)$,

$$A^8 = \begin{bmatrix} 766 & -765 \\ 510 & -509 \end{bmatrix}.$$

This result was also checked by eightfold exact matrix multiplication.

Practice problem 5.3.P2: Use the supplied eigenvectors

Multiplying by $A = \left[\begin{smallmatrix} -3 & 12 \\ -2 & 7 \end{smallmatrix}\right]$ gives $A(3,1)^T = (3,1)^T$ and $A(2,1)^T = 3(2,1)^T$. The eigenvalues are therefore 1 and 3, in that order. Take $P = \left[\begin{smallmatrix} 3 & 2 \\ 1 & 1 \end{smallmatrix}\right]$ and $D = \operatorname{diag}(1,3)$. Since $\det P = 1 \neq 0$ and $AP = PD$, we obtain $A = PDP^{-1}$.

Practice problem 5.3.P3: Count eigenspace dimensions

Yes. The eigenspace for 3 supplies two independent vectors. The distinct eigenvalues 5 and -2 each supply at least one nonzero eigenvector. Combining bases from distinct eigenspaces gives independent vectors, hence at least $2 + 1 + 1 = 4$ independent vectors in $\mathbb{R}^4$. There cannot be more than four. Thus they form an eigenvector basis and the matrix is diagonalizable.

Exercise 5.3.1: Cancel the inner change-of-basis factors

What we must show. Compute A^4 from the given $A = PDP^{-1}$. Associativity gives $(PDP^{-1})^4 = PD(P^{-1}P)D(P^{-1}P)D(P^{-1}P)DP^{-1} = PD^4P^{-1}$. Here

$$P = \begin{bmatrix} 5 & 7 \\ 2 & 3 \end{bmatrix}, \quad D^4 = \begin{bmatrix} 16 & 0 \\ 0 & 1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 3 & -7 \\ -2 & 5 \end{bmatrix}.$$

Multiplying these three displayed matrices gives

$$A^4 = \begin{bmatrix} 226 & -525 \\ 90 & -209 \end{bmatrix}.$$

The calculation involves taking fourth powers of the diagonal entries of D, not taking fourth powers entry by entry in A.

Exercise 5.3.2: Cancel the inner change-of-basis factors

What we must show. Compute A^4 from the given $A = PDP^{-1}$. Associativity gives $(PDP^{-1})^4 = PD(P^{-1}P)D(P^{-1}P)D(P^{-1}P)DP^{-1} = PD^4P^{-1}$. Here

$$P = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}, \quad D^4 = \begin{bmatrix} 1 & 0 \\ 0 & 81 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix}.$$

Multiplying these three displayed matrices gives

$$A^4 = \begin{bmatrix} 321 & -160 \\ 480 & -239 \end{bmatrix}.$$

The calculation involves taking fourth powers of the diagonal entries of D, not taking fourth powers entry by entry in A.

Exercise 5.3.3: An arbitrary power of a parameterized triangular matrix

Use the supplied factorization with $P = \left[\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right]$, $D = \operatorname{diag}(a, b)$, and $P^{-1} = \left[\begin{smallmatrix} 1 & 0 \\ -2 & 1 \end{smallmatrix}\right]$. For every positive integer k,

$$A^k = PD^kP^{-1} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} a^k & 0 \\ 0 & b^k \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} a^k & 0 \\ 2(a^k - b^k) & b^k \end{bmatrix}.$$

There is no division by $a - b$, so this remains true when $a = b$ or one parameter is zero.

Exercise 5.3.4: An arbitrary power with negative eigenvalues

Put $\alpha = (-3)^k$ and $\beta = (-2)^k$. The supplied matrices give

$$A^k = \begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix} \begin{bmatrix} -1 & 2 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} -3\alpha + 4\beta & 6\alpha - 6\beta \\ -2\alpha + 2\beta & 4\alpha - 3\beta \end{bmatrix}.$$

The signs in $(-3)^k$ and $(-2)^k$ depend on the parity of k and must remain inside parentheses. Setting $k = 1$ recovers the original matrix, a useful quick check.

Exercise 5.3.5: Read eigenspaces from an existing factorization

Read matching columns and diagonal entries in the supplied factorization. If p_j is column j of P, then $Ap_j = d_{jj}p_j$. Here

$$P = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

For $\lambda = 2$, a basis is $\{\begin{bmatrix}-1\\1\\-1\end{bmatrix}\}$. For $\lambda = 3$, a basis is $\{\begin{bmatrix}1\\-1\\0\end{bmatrix}, \begin{bmatrix}0\\-1\\1\end{bmatrix}\}$. Because P is invertible, its columns are independent. All columns at a repeated diagonal value are needed; retaining only one of them would lose an eigenspace direction.

Exercise 5.3.6: Read eigenspaces from an existing factorization

Read matching columns and diagonal entries in the supplied factorization. If p_j is column j of P, then $Ap_j = d_{jj}p_j$. Here

$$P = \begin{bmatrix}3 & 0 & -1\\0 & 1 & -3\\1 & 0 & 0\end{bmatrix}, \qquad D = \begin{bmatrix}3 & 0 & 0\\0 & 4 & 0\\0 & 0 & 3\end{bmatrix}.$$

For $\lambda = 3$, a basis is $\{\begin{bmatrix}3\\0\\1\end{bmatrix}, \begin{bmatrix}-1\\-3\\0\end{bmatrix}\}$. For $\lambda = 4$, a basis is $\{\begin{bmatrix}0\\1\\0\end{bmatrix}\}$. Because P is invertible, its columns are independent. All columns at a repeated diagonal value are needed; retaining only one of them would lose an eigenspace direction.

Exercise 5.3.7: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix}1 & 0\\6 & -1\end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-1)(t+1).$$

Find the eigenspace for $\lambda = -1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-1)I) = \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-1} = \operatorname{Span}\left\{\begin{bmatrix}0\\1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-1)v$ for each listed vector.

Find the eigenspace for $\lambda = 1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (1)I) = \begin{bmatrix} 1 & -\frac{1}{3} \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_1 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (1)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 3 \end{bmatrix}, \qquad D = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The exact checks are $\det P = -1 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.8: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 3 & 2 \\ 0 & 3 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 3)^2.$$

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Columns 1 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 = 1$. This is smaller than 2, so no eigenvector basis exists. Hence A is not diagonalizable. Repeating or rescaling one of these vectors cannot supply a missing independent direction.

Exercise 5.3.9: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-3)^2.$$

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 = 1$. This is smaller than 2, so no eigenvector basis exists. Hence A is not diagonalizable. Repeating or rescaling one of these vectors cannot supply a missing independent direction.

Exercise 5.3.10: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 5)(t + 2).$$

Find the eigenspace for $\lambda = -2$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-2)I) = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-2} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-2)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & -\frac{3}{4} \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span}\left\{ \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 3 \\ -1 & 4 \end{bmatrix}, \qquad D = \begin{bmatrix} -2 & 0 \\ 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = 7 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.11: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 2 & 1 & 2 \\ 3 & 3 & 2 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-5)(t+1)^2.$$

Find the eigenspace for $\lambda = -1$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-1)I) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-1} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-1)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & -\frac{1}{3} \\ 0 & 1 & -\frac{2}{3} \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 = 3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ 0 & -1 & 3 \end{bmatrix}, \qquad D = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = 6 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.12: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 5)\,(t - 2)^2.$$

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (2)I) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_2 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (2)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \text{Span}\left\{\begin{bmatrix}1\\1\\1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 = 3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix}1 & 1 & 1\\ -1 & 0 & 1\\ 0 & -1 & 1\end{bmatrix}, \qquad D = \begin{bmatrix}2 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 5\end{bmatrix}.$$

The exact checks are $\det P = 3 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.13: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix}2 & 2 & -1\\ 1 & 3 & -1\\ -1 & -2 & 2\end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 5)\,(t - 1)^2.$$

Find the eigenspace for $\lambda = 1$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\text{rref}(A - (1)I) = \begin{bmatrix}1 & 2 & -1\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_1 = \text{Span}\left\{\begin{bmatrix}2\\-1\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (1)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 = 3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix}, \qquad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = -4 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.14: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 2 & 0 & -2 \\ 1 & 3 & 2 \\ 0 & 0 & 3 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 3)^2 (t - 2).$$

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 1. Solve $(A-\lambda I)x=0$; row reduction gives

$$\operatorname{rref}(A-(2)I)=\begin{bmatrix}1&1&0\\0&0&1\\0&0&0\end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_2=\operatorname{Span}\left\{\begin{bmatrix}1\\-1\\0\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av=(2)v$ for each listed vector.

Find the eigenspace for $\lambda=3$.

The algebraic multiplicity is 2. Solve $(A-\lambda I)x=0$; row reduction gives

$$\operatorname{rref}(A-(3)I)=\begin{bmatrix}1&0&2\\0&0&0\\0&0&0\end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3=\operatorname{Span}\left\{\begin{bmatrix}0\\1\\0\end{bmatrix},\ \begin{bmatrix}2\\0\\-1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av=(3)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1+2=3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P=\begin{bmatrix}1&0&2\\-1&1&0\\0&0&-1\end{bmatrix},\qquad D=\begin{bmatrix}2&0&0\\0&3&0\\0&0&3\end{bmatrix}.$$

The exact checks are $\det P=-1\neq 0$ and $AP=PD$. Multiplying the latter by P^{-1} on the right proves $A=PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.15: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A=\begin{bmatrix}0&-1&-1\\1&2&1\\-1&-1&0\end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = t\,(t-1)^2\,.$$

Find the eigenspace for $\lambda = 0$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (0)I) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_0 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (0)v$ for each listed vector.

Find the eigenspace for $\lambda = 1$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (1)I) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_1 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (1)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 2 = 3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}, \qquad D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The exact checks are $\det P = 1 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.16: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 5 & -2 \\ 1 & 3 & 1 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = t\left(t^2 - 7t + 16\right).$$

Find the eigenspace for $\lambda = 0$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (0)I) = \begin{bmatrix} 1 & 0 & -11 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_0 = \operatorname{Span}\left\{ \begin{bmatrix} 11 \\ -4 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (0)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 = 1$. This is smaller than 3, so no eigenvector basis exists. Hence A is not diagonalizable. Repeating or rescaling one of these vectors cannot supply a missing independent direction. The remaining quadratic factor is $t^2 - 7t + 16$, whose discriminant is $49 - 64 = -15 < 0$. Thus its other eigenvalues are nonreal. They do not supply real eigenvectors for a basis of $\mathbb{R}^3$; complex diagonalization is a different question introduced in Section 5.5.

Exercise 5.3.17: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-2)^3 .$$

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 3. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (2)I) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} .$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_2 = \operatorname{Span}\left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} .$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (2)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 = 1$. This is smaller than 3, so no eigenvector basis exists. Hence A is not diagonalizable. Repeating or rescaling one of these vectors cannot supply a missing independent direction.

Exercise 5.3.18: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 2 & -2 & -2 \\ 3 & -3 & -2 \\ 2 & -2 & -2 \end{bmatrix} .$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = t\,(t+1)\,(t+2) .$$

Find the eigenspace for $\lambda = -2$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-2)I) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} .$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-2} = \text{Span}\left\{\begin{bmatrix}1\\1\\1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-2)v$ for each listed vector.

Find the eigenspace for $\lambda = -1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\text{rref}(A - (-1)I) = \begin{bmatrix}1 & 0 & -1\\0 & 1 & -\frac{1}{2}\\0 & 0 & 0\end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-1} = \text{Span}\left\{\begin{bmatrix}2\\1\\2\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-1)v$ for each listed vector.

Find the eigenspace for $\lambda = 0$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\text{rref}(A - (0)I) = \begin{bmatrix}1 & -1 & 0\\0 & 0 & 1\\0 & 0 & 0\end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{0} = \text{Span}\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (0)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 + 1 = 3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix}1 & 2 & 1\\1 & 1 & 1\\1 & 2 & 0\end{bmatrix}, \qquad D = \begin{bmatrix}-2 & 0 & 0\\0 & -1 & 0\\0 & 0 & 0\end{bmatrix}.$$

The exact checks are $\det P = 1 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.19: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 5 & -3 & 0 & 9 \\ 0 & 3 & 1 & -2 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-5)(t-3)(t-2)^2.$$

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (2)I) = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 3, 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_2 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (2)v$ for each listed vector.

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 1 & -\frac{3}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 1 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 + 1 = 4$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 1 & 3 & 1 \\ 1 & -2 & 2 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}, \qquad D = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = -2 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.20: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 3 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-3)^2 (t-2)^2.$$

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (2)I) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_2 = \operatorname{Span}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (2)v$ for each listed vector.

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 = 3$. This is smaller than 4, so no eigenvector basis exists. Hence A is not diagonalizable. Repeating or rescaling one of these vectors cannot supply a missing independent direction.

Exercise 5.3.21: Diagonalization criteria and constructions

(a) False.

The condition must say that D is *diagonal*. Without that, every matrix has the factorization $A = IAI^{-1}$, including a nondiagonalizable Jordan block.

(b) True.

A basis consists of n independent vectors; the Diagonalization Theorem applies to that eigenvector basis.

(c) False.

Counting roots with multiplicities does not count independent eigenvectors. For instance $\left[\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right]$ has characteristic polynomial $(t-1)^2$ but only the one-dimensional eigenspace $\text{Span}\{(1,0)\}$.

(d) False.

A diagonal matrix can have a zero diagonal entry. The zero matrix is diagonalizable but not invertible.

Exercise 5.3.22: Diagonalization criteria and constructions

(a) False.

The vectors must be linearly independent. One eigenvector and its nonzero scalar multiples give arbitrarily many eigenvectors but only one direction.

(b) False.

I_n is diagonalizable and has only the eigenvalue 1 when $n > 1$.

(c) True.

Taking column j of $AP = PD$ gives $Ap_j = d_{jj}p_j$. If $p_j \neq 0$, this is exactly an eigenvector equation; zero columns are excluded by the definition.

(d) False.

The Jordan block $\left[\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right]$ has determinant 1 and is invertible, but its eigenspace has dimension 1, so it is not diagonalizable.

Exercise 5.3.23: Diagonalization criteria and constructions

Yes. Choose a basis of three vectors from the first eigenspace and a basis of two from the other. Eigenvectors from different eigenspaces combine independently, so the union has $3+2 = 5$ independent vectors in $\mathbb{R}^5$. It is an eigenvector basis, giving diagonalizability.

Exercise 5.3.24: Diagonalization criteria and constructions

No. The dimensions of the only two eigenspaces add to $1 + 1 = 2$, which is less than 3. Every eigenvector belongs to one of those spaces, so one cannot obtain a third independent eigenvector. Thus the Diagonalization Theorem fails.

Exercise 5.3.25: Diagonalization criteria and constructions

No: under these assumptions the matrix must be diagonalizable. The two known dimensions are 1 and 2. The third eigenspace has at least dimension 1, giving at least $1 + 2 + 1 = 4$ independent

eigenvectors from the three distinct eigenspaces. Since the ambient space has dimension 4, this is exactly a basis.

Exercise 5.3.26: Diagonalization criteria and constructions

Yes, failure is possible. The known dimensions add to $2 + 3 = 5$, and the third can have dimension 1 or 2. If it has dimension 1, the total is $6 < 7$. For a concrete example take

$$A = \operatorname{diag}(0, 0, 1, 1, 1, J), \qquad J = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

The eigenspace dimensions for $0, 1, 2$ are $2, 3, 1$, respectively, so this 7×7 matrix is not diagonalizable. If the last block were $2I_2$, the third dimension would instead be 2 and the matrix would be diagonalizable. The supplied data do not decide which occurs.

Exercise 5.3.27: Diagonalization criteria and constructions

Write $A = PDP^{-1}$. Invertibility of A implies no diagonal entry of D is zero. Inverting the product in reverse order gives $A^{-1} = PD^{-1}P^{-1}$, where D^{-1} is again diagonal. Thus A^{-1} is diagonalizable, and it is invertible because its inverse is A. Equivalently, $Av = \lambda v$ implies $A^{-1}v = \lambda^{-1}v$ for the same eigenvector basis.

Exercise 5.3.28: Diagonalization criteria and constructions

The hypothesis means $A = PDP^{-1}$ with D diagonal. Transpose this equality:

$$A^T = (P^{-1})^T D P^T = QDQ^{-1}, \qquad Q = (P^{-1})^T.$$

The matrix Q is invertible, and its columns therefore form n independent eigenvectors of A^T. The transpose generally does not have the *same* individual eigenvectors; the point is that it has its own eigenvector basis.

Exercise 5.3.29: Diagonalization criteria and constructions

In Example 2, $A = \left[\begin{smallmatrix} 7 & 2 \\ -4 & 1 \end{smallmatrix}\right]$, with eigenvectors $(1, -1)$ for 5 and $(1, -2)$ for 3. The new $D_1 = \operatorname{diag}(3, 5)$ reverses the eigenvalue order, so reverse the columns as well:

$$P_1 = \begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix}.$$

Its determinant is $1 \neq 0$, and multiplication verifies $AP_1 = P_1 D_1$. Thus $A = P_1 D_1 P_1^{-1}$. Reordering eigenvalues without reordering eigenvectors would generally be wrong.

Exercise 5.3.30: Diagonalization criteria and constructions

Keep $D = \operatorname{diag}(5, 3)$ from Example 2, but rescale its first eigenvector. For instance,

$$P_2 = \begin{bmatrix} 2 & 1 \\ -2 & -2 \end{bmatrix}$$

has columns $2(1,-1)$ and $(1,-2)$ and determinant $-2 \neq 0$. It differs from the displayed P, but $AP_2 = P_2D$ still holds, so $A = P_2DP_2^{-1}$. Any nonzero rescaling of a basis eigenvector gives another valid factorization with that eigenvalue order.

Exercise 5.3.31: Diagonalization criteria and constructions

Take $A = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$. Its determinant is 1, so it is invertible and certainly nonzero. Its only eigenvalue is 1, and $(A - I)x = 0$ says $x_2 = 0$, leaving only a one-dimensional eigenspace. Therefore it is not diagonalizable.

Exercise 5.3.32: Diagonalization criteria and constructions

Take $A = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right]$. It is nondiagonal and has determinant 0, so is not invertible. Its distinct eigenvalues are 1 and 0; corresponding independent eigenvectors are $(1,0)$ and $(-1,1)$. Thus with $P = \left[\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right]$ and $D = \operatorname{diag}(1,0)$, we have $AP = PD$ and $A = PDP^{-1}$.

Exercise 5.3.33: Exact computer-assisted diagonalization

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 9 & -4 & -2 & -4 \\ -56 & 32 & -28 & 44 \\ -14 & -14 & 6 & -14 \\ 42 & -33 & 21 & -45 \end{bmatrix}.$$

This is a matrix-program exercise; the eigenvalue and null-space calculations below were carried out with exact rational arithmetic, then checked by multiplication.

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-13)^2\,(t+12)^2\,.$$

Find the eigenspace for $\lambda = -12$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-12)I) = \begin{bmatrix} 1 & 0 & -\frac{2}{7} & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 3, 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-12} = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ 7 \\ 7 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-12)v$ for each listed vector.

Find the eigenspace for $\lambda = 13$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (13)I) = \begin{bmatrix} 1 & 0 & \frac{1}{2} & -\frac{1}{3} \\ 0 & 1 & 0 & \frac{4}{3} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 3, 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{13} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -4 \\ 0 \\ 3 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (13)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 2 = 4$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 2 & 0 & 1 & 1 \\ 7 & 1 & 0 & -4 \\ 7 & 0 & -2 & 0 \\ 0 & -1 & 0 & 3 \end{bmatrix}, \qquad D = \begin{bmatrix} -12 & 0 & 0 & 0 \\ 0 & -12 & 0 & 0 \\ 0 & 0 & 13 & 0 \\ 0 & 0 & 0 & 13 \end{bmatrix}.$$

The exact checks are $\det P = 25 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. A reproducible exact command pattern is:

```
for lam,multiplicity,basis in A.eigenvects():
    print(lam,multiplicity,(A-lam*eye(A.rows)).rref())
    print(basis)
```

Here `A` is the displayed matrix and `Matrix, eye` are imported from `sympy`. The printed bases may be rescaled without changing their spans.

Exercise 5.3.34: Exact computer-assisted diagonalization

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 4 & -9 & -7 & 8 & 2 \\ -7 & -9 & 0 & 7 & 14 \\ 5 & 10 & 5 & -5 & -10 \\ -2 & 3 & 7 & 0 & 4 \\ -3 & -13 & -7 & 10 & 11 \end{bmatrix}.$$

This is a matrix-program exercise; the eigenvalue and null-space calculations below were carried out with exact rational arithmetic, then checked by multiplication.

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-5)^3 (t+2)^2 .$$

Find the eigenspace for $\lambda = -2$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-2)I) = \begin{bmatrix} 1 & 0 & 0 & \frac{2}{5} & -\frac{3}{5} \\ 0 & 1 & 0 & -\frac{7}{5} & -\frac{2}{5} \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} .$$

Columns 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-2} = \operatorname{Span} \left\{ \begin{bmatrix} 2 \\ -7 \\ 5 \\ -5 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \\ -5 \\ 0 \\ 5 \end{bmatrix} \right\} .$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-2)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 3. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & -2 & 1 & -2 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} .$$

Columns 3, 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span} \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\} .$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 3. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 3 = 5$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 2 & 3 & 2 & 1 & 2 \\ -7 & 7 & -1 & -1 & 0 \\ 5 & -5 & 1 & 0 & 0 \\ -5 & 0 & 0 & -1 & 0 \\ 0 & 5 & 0 & 0 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} -2 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = 35 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. A reproducible exact command pattern is:

```
for lam,multiplicity,basis in A.eigenvects():
    print(lam,multiplicity,(A-lam*eye(A.rows)).rref())
    print(basis)
```

Here `A` is the displayed matrix and `Matrix, eye` are imported from `sympy`. The printed bases may be rescaled without changing their spans.

Exercise 5.3.35: Exact computer-assisted diagonalization

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 13 & -12 & 9 & -15 & 9 \\ 6 & -5 & 9 & -15 & 9 \\ 6 & -12 & -5 & 6 & 9 \\ 6 & -12 & 9 & -8 & 9 \\ -6 & 12 & 12 & -6 & -2 \end{bmatrix}.$$

This is a matrix-program exercise; the eigenvalue and null-space calculations below were carried out with exact rational arithmetic, then checked by multiplication.

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 7)^3 (t + 14)^2.$$

Find the eigenspace for $\lambda = -14$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-14)I) = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-14} = \text{Span}\left\{\begin{bmatrix}1\\1\\0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\\0\\-1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-14)v$ for each listed vector.

Find the eigenspace for $\lambda = 7$.

The algebraic multiplicity is 3. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\text{rref}(A - (7)I) = \begin{bmatrix}1 & -2 & 0 & -1 & \frac{3}{2}\\0 & 0 & 1 & -1 & 0\\0 & 0 & 0 & 0 & 0\\0 & 0 & 0 & 0 & 0\\0 & 0 & 0 & 0 & 0\end{bmatrix}.$$

Columns 2, 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_7 = \text{Span}\left\{\begin{bmatrix}2\\1\\0\\0\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\0\\0\\-2\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 3. Direct multiplication checks $Av = (7)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 3 = 5$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix}1 & 0 & 2 & 1 & 3\\1 & 0 & 1 & 0 & 0\\0 & 1 & 0 & 1 & 0\\1 & 0 & 0 & 1 & 0\\0 & -1 & 0 & 0 & -2\end{bmatrix}, \qquad D = \begin{bmatrix}-14 & 0 & 0 & 0 & 0\\0 & -14 & 0 & 0 & 0\\0 & 0 & 7 & 0 & 0\\0 & 0 & 0 & 7 & 0\\0 & 0 & 0 & 0 & 7\end{bmatrix}.$$

The exact checks are $\det P = -7 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. A reproducible exact command pattern is:

```
for lam,multiplicity,basis in A.eigenvects():
    print(lam,multiplicity,(A-lam*eye(A.rows)).rref())
    print(basis)
```

Here `A` is the displayed matrix and `Matrix, eye` are imported from `sympy`. The printed bases may be rescaled without changing their spans.

Exercise 5.3.36: Exact computer-assisted diagonalization

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 24 & -6 & 2 & 6 & 2 \\ 72 & 51 & 9 & -99 & 9 \\ 0 & -63 & 15 & 63 & 63 \\ 72 & 15 & 9 & -63 & 9 \\ 0 & 63 & 21 & -63 & -27 \end{bmatrix}.$$

This is a matrix-program exercise; the eigenvalue and null-space calculations below were carried out with exact rational arithmetic, then checked by multiplication.

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 36)^2 (t - 24) (t + 48)^2 .$$

Find the eigenspace for $\lambda = -48$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-48)I) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-48} = \operatorname{Span}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-48)v$ for each listed vector.

Find the eigenspace for $\lambda = 24$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (24)I) = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{24} = \text{Span}\left\{\begin{bmatrix}1\\1\\0\\1\\0\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (24)v$ for each listed vector.

Find the eigenspace for $\lambda = 36$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\text{rref}(A - (36)I) = \begin{bmatrix}1 & 0 & 0 & -1 & \frac{1}{3}\\0 & 1 & 0 & 0 & -1\\0 & 0 & 1 & -3 & 0\\0 & 0 & 0 & 0 & 0\\0 & 0 & 0 & 0 & 0\end{bmatrix}.$$

Columns 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{36} = \text{Span}\left\{\begin{bmatrix}1\\0\\3\\1\\0\end{bmatrix}, \begin{bmatrix}1\\-3\\0\\0\\-3\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (36)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 + 2 = 5$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix}0 & 0 & 1 & 1 & 1\\1 & 0 & 1 & 0 & -3\\0 & 1 & 0 & 3 & 0\\1 & 0 & 1 & 1 & 0\\0 & -1 & 0 & 0 & -3\end{bmatrix}, \qquad D = \begin{bmatrix}-48 & 0 & 0 & 0 & 0\\0 & -48 & 0 & 0 & 0\\0 & 0 & 24 & 0 & 0\\0 & 0 & 0 & 36 & 0\\0 & 0 & 0 & 0 & 36\end{bmatrix}.$$

The exact checks are $\det P = -12 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. A reproducible exact command pattern is:

```
for lam,multiplicity,basis in A.eigenvects():
    print(lam,multiplicity,(A-lam*eye(A.rows)).rref())
    print(basis)
```

Here `A` is the displayed matrix and `Matrix, eye` are imported from `sympy`. The printed bases may be rescaled without changing their spans.